Cryptography

Lecture 5: Attacking Block Ciphers

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Learning Objectives

By the end of this lecture, you will be able to:

- Analyze the key size vulnerabilities of DES
- Understand meet-in-the-middle attacks and their complexity
- Explain differential cryptanalysis techniques and requirements
- Describe linear cryptanalysis and bias exploitation
- Compare the effectiveness of different cryptanalytic approaches

Weaknesses of DES

DES Key Size Problem

Historical context: Even in 1977, 56-bit key size was considered low

NSA involvement: “Encouraged” reduction from 128 to 56 bits

Key space analysis:

- Total possible keys: $2^{\{56\}} \approx 7.2 \times 10^{\{16\}}$
- Expected search effort: $2^{\{55\}}$ keys on average
- Brute force becomes increasingly feasible over time

Critical weakness: Small key space makes exhaustive search practical

Historical DES Breaking Attempts

Timeline of proposed/actual attacks:

- 1977: Diffie & Hellman - $20 million machine, 1 day
- 1993: Wiener - $1 million machine, 7 hours
- 1997: RSA challenge - 78,000 PCs, distributed computing
- 1998: EFF machine - $250,000, 2 days
- 2006: COPACOBANA - $10,000, less than 1 week

Key observation: Cost and time requirements decreased exponentially

Key Size Attack Analysis

DES (56-bit keys):

- Search space: $2^{\{56\}}$ keys
- Expected effort: $2^{\{55\}}$ keys
- At $2^{\{30\}}$ keys/second: 1 year to break

AES-128 (128-bit keys):

- Search space: $2^{\{128\}}$ keys
- Expected effort: $2^{\{127\}}$ keys
- At $2^{\{60\}}$ keys/second: $10^{\{13\}}$ years

AES-256 (256-bit keys):

- Requires $2^{\{128\}}$ times more power than AES-128
- At $10^{\{18\}}$ keys/second: $10^{\{51\}}$ years

Perspective: $10^{\{13\}}$ years is 1000× longer than age of universe

Meet-in-the-Middle Attacks

Double Encryption Vulnerability

Naive approach: Use two keys for double encryption

$$C = e(e(M, K_1), K_2)$$

Expected security: $2k$ bits (where k is single key length)

Actual security: Much less due to meet-in-the-middle attack

Key insight: Double encryption does not double security!

Meet-in-the-Middle Attack Method

Given: Known plaintext-ciphertext pair (M, C)

Attack strategy:

1. Compute table of $e(M, K)$ for all 2^k possible keys K
2. For each possible key K', compute $d(C, K')$
3. Check for matches: $e(M, K) = d(C, K')$
4. When match found: $e(e(M, K), K') = C$

Complexity:

- Time: $2^{\{k+1\}}$ operations (not $2^{\{2k\}}$!)
- Space: 2^k table entries

Result: Attack complexity is $O(2^k)$, not $O(2^{\{2k\}})$

Triple DES and Meet-in-the-Middle

Triple DES construction:

$$C = e(d(e(M, K_1), K_2), K_3)$$

Meet-in-the-middle on 3DES:

1. Enumerate $d(e(M, K_1), K_2)$ for all K_1, K_2 ($2^{\{2k\}}$ options)
2. Check matches with $d(C, K_3)$ for all K_3 (2^k options)

Total complexity: $2^{\{2k\}} + 2^k$ operations

For DES: $2^{\{112\}} + 2^{\{56\}} \ll 2^{\{168\}}$ (much less than brute force)

General Meet-in-the-Middle Principles

Applicability: Any cipher with chained encryption operations

Key requirements:

- Access to known plaintext-ciphertext pairs
- Ability to compute intermediate values
- Sufficient storage for precomputed tables

Defense strategies:

- Use constructions that resist this attack
- Ensure intermediate values cannot be easily computed
- Design key schedules that create dependencies

Lesson: Simply chaining encryptions doesn’t proportionally increase security

Differential Cryptanalysis

Differential Cryptanalysis Overview

Definition: Chosen-plaintext attack comparing XOR of inputs to XOR of outputs

Core concept: Analyze how differences propagate through cipher

Attack requirements:

- Large number of chosen plaintext pairs
- Fixed input difference $x' = x \oplus x^*$
- Access to corresponding ciphertext pairs

Historical note:

- Discovered by Biham & Shamir (late 1980s)
- IBM & NSA knew about it since mid-1970s
- DES was designed to resist this attack

Goal: Find input differences that cause predictable output differences

Differential Analysis: S-box Example

Example S-box: 4-bit input → 4-bit output

Process:

1. Consider all possible input differentials (16 possibilities)
2. For each input differential, count output differentials
3. Look for biased patterns

Key finding: Input differential `1111` → Output differential `1010` occurs 8 times

Expected behavior: Random function would show each output differential 1 time

Security implication: This bias can be exploited in attacks

Weakness: Predictable differential behavior reveals internal structure

Differential Attack on Cipher Structure

Target: Substitution-permutation network with 3 subkeys

Brute force complexity: $2^{\{192\}}$ (three 64-bit subkeys)

Meet-in-the-middle: $2^{\{128\}}$ operations

Differential attack strategy:

1. Find input differential $D_{\{\text{in}\}}$ causing predictable $D_{\{\text{mid}\}}$
2. Use bias: $D_{\{\text{mid}\}}$ occurs with probability $2^{\{-50\}}$ (vs. $2^{\{-64\}}$ random)
3. Generate $L = 2^{\{54\}}$ pairs with differential $D_{\{\text{in}\}}$
4. For each guess of K_3: decrypt and count $D_{\{\text{mid}\}}$ occurrences
5. Correct K_3 produces multiple matches; wrong K_3 produces ≤1

Differential Attack Complexity

Attack steps:

1. Generate $2^{\{54\}}$ chosen plaintext pairs
2. Test $2^{\{64\}}$ possible values of K_3
3. For each K_3 guess, decrypt $2^{\{54\}}$ ciphertext pairs
4. Use meet-in-the-middle for remaining subkeys

Total complexity: $2^{\{64\}} \times 2^{\{54\}} \times 2 = 2^{\{119\}}$ decryptions

Improvement: $2^{\{119\}} < 2^{\{128\}}$ (meet-in-the-middle) $\ll 2^{\{192\}}$ (brute force)

Key insight: Exploiting S-box biases reduces attack complexity significantly

Linear Cryptanalysis

Linear Cryptanalysis Overview

Definition: Known-plaintext attack exploiting linear approximations

Key advantage: Only requires known (not chosen) plaintext-ciphertext pairs

Core principle: Exploit bias in linear equations relating input and output bits

Historical context:

- Introduced by Matsui (1994)
- Successfully applied to DES the same year
- Required $2^{\{43\}}$ plaintext-ciphertext pairs

Practical advantage: More realistic than differential cryptanalysis

- Works with intercepted communications
- Doesn’t require ability to choose plaintexts

Linear Functions and Security

Linear encryption example:

$$c_{\{15\}} = p_1 \oplus p_3 \oplus p_{\{18\}} \oplus p_{\{22\}} \oplus k_2 \oplus k_7 \oplus k_{\{10\}}$$

Security requirement: Ciphers should be as far from linear as possible

Ideal behavior: Any linear equation should hold with probability ≈ 1/2

S-box vulnerability: If linear equations have significant bias, attack possible

Bias definition: Probability deviates significantly from 1/2

Critical: S-boxes must minimize bias in linear approximations

Linear Analysis: S-box Example

Same 4-bit S-box from differential analysis

Linear equation analysis:

- Input bits: x_1, x_2, x_3, x_4
- Output bits: y_1, y_2, y_3, y_4

Example bias: Equation $y_1 = x_1 \oplus x_3 \oplus x_4$

- True for 14 out of 16 inputs
- Expected for random function: 8 out of 16
- Bias: $\frac{14}{16} - \frac{1}{2} = \frac{3}{8}$ (very large!)

Conclusion: This S-box has poor linearity properties

Matsui's Linear Attack on DES

Matsui's 14-round linear relation in DES:

$$m_{R[17,18,24]} \oplus c_{L[7,18,24,29]} \oplus c_{R[15]} = k[S_e]$$

Bias: $\frac{1}{2} + 2^{\{-21\}}$ (holds with this probability)

Key insight: This relation involves specific bit positions and holds with detectable bias over many plaintext-ciphertext pairs

Challenge: Need to extend this to full 16-round DES by incorporating first and last rounds

Extended Linear Attack on Full DES

Extension to 16 rounds: Incorporate first and last rounds

$$(m_L \oplus F(k_1, m_R))[17, 18, 24] \oplus c_{R[7,18,24,29]} \oplus \left(c_L \oplus F\left(k_{\{16\}}, c_R\right)\right)[15] = k[S'_e]$$

Key observation:

- Bits 17,18,24 of $F(k_1, m_R)$ depend on only 6 bits of k_1
- Bit 15 of $F\left(k_{\{16\}}, c_R\right)$ depends on only 6 bits of $k_{\{16\}}$
- Total dependency: only 12 key bits!

Attack strategy: Test all $2^{\{12\}}$ combinations of these 12 key bits

- Correct key shows high bias
- Wrong keys show low bias

DES Linear Attack: Complexity and Impact

Resource requirements:

- $2^{\{43\}}$ known plaintext-ciphertext pairs
- No chosen plaintext capability needed
- All pairs encrypted with same unknown key

Computational complexity:

- Test $2^{\{12\}} = 4096$ key bit combinations
- For each: evaluate linear equation on $2^{\{43\}}$ pairs
- Much more efficient than brute force $2^{\{56\}}$ key search

Historical significance:

- First successful linear attack on DES (Matsui, 1994)
- Demonstrated practical cryptanalytic technique
- Influenced AES design to resist linear attacks

Limitation: $2^{\{43\}}$ pairs still impractical for real-world impact

Defense Strategies

Against brute force attacks:

- Use sufficiently large key sizes (≥128 bits)
- Regular key rotation

Against meet-in-the-middle:

- Avoid simple chaining of encryptions
- Use key-dependent operations throughout

Against differential cryptanalysis:

- Design S-boxes with uniform differential properties
- Use sufficient number of rounds
- Ensure good diffusion between rounds

Against linear cryptanalysis:

- Minimize bias in S-box linear approximations
- Use nonlinear operations throughout
- Test for linear characteristics during design

Modern approach: AES designed with resistance to all these attacks

Advanced Encryption Standard (AES)

Introduction to AES

Background: Advanced Encryption Standard introduced in 2002 to replace DES

Selection process: Open competition with full transparency

- All algorithms required to explain design details
- Contrast with DES where some details were kept secret
- Multiple rounds of evaluation and public analysis

Key improvements over DES:

- Larger key sizes (128, 192, 256 bits)
- Designed to resist known cryptanalytic attacks
- Algebraically derived S-boxes with proven properties

Goal: Create a modern, secure, and efficient block cipher standard

AES Structure: Substitution-Permutation Network

Architecture: AES uses a substitution-permutation network (not Feistel)

Core components:

- Substitutions (S-boxes) for nonlinearity
- Permutations (linear operations) for diffusion
- Subkey addition for confusion

Key difference from Feistel:

- S-boxes must be bijections (invertible)
- Required for decryptability
- Operates on entire block simultaneously

Structure: Alternating layers of substitution and permutation with key addition

AES Parameters and Block Structure

Block size: 128 bits (fixed)

- Represented as 4×4 array of bytes
- Each cell contains 8 bits

Key sizes: 128, 192, or 256 bits

Number of rounds:

- AES-128: 10 rounds
- AES-192: 12 rounds
- AES-256: 14 rounds

Key Size	Rounds	Security Level
128 bits	10	128 bits
192 bits	12	192 bits
256 bits	14	256 bits

AES Round Operations

Each AES round consists of four operations:

1. SubBytes: S-box transformation (nonlinear substitution)
2. ShiftRows: Cyclically shift rows of state array
3. MixColumns: Linear mixing of columns
4. AddRoundKey: XOR with round subkey

Key insights:

- Operations 1-3 are fixed (algorithm-dependent)
- Only AddRoundKey uses the secret key
- SubBytes provides nonlinearity (critical for security)
- ShiftRows and MixColumns provide diffusion

Design principle: Separate confusion (SubBytes) from diffusion (ShiftRows/MixColumns)

AES Key Schedule

Purpose: Generate round subkeys from master key

Requirements: Each round needs 128-bit subkey

Generation process:

- Linear transformations of previous subkeys
- Application of S-box for nonlinearity
- Round constants for each round

Security goals:

- Ensure subkeys appear independent
- Prevent related-key attacks
- Provide sufficient key material for all rounds

Key expansion: Master key expanded to (rounds + 1) × 128 bits

AES S-box Design: Mathematical Foundation

Mathematical definition: S-box is function $\{0,1\}^n \rightarrow \{0,1\}^m$

Component view: Tuple of m Boolean functions $(s_1, ..., s_m)$

Security-critical parameters:

- Nonlinearity
- Balancedness
- Resiliency
- Correlation immunity
- Algebraic degree

AES approach: Algebraically derived S-boxes

- Based on multiplicative inverse over $\mathrm{GF}(2^8)$
- Mathematically proven good nonlinearity properties
- Transparent design process

AES vs DES: S-box Comparison

Aspect	DES	AES
S-box source	Lookup tables	Algebraic derivation
Justification	None given	Mathematical proof
Size	6→4 bits	8→8 bits
Invertibility	Not required	Must be bijection
Transparency	Design secret	Fully public

Key advantage: AES S-boxes have provable resistance to linear and differential cryptanalysis

Result: More robust against known attack techniques

AES Security Status

Cryptanalytic resistance:

- Designed to resist linear cryptanalysis
- Resilient to differential cryptanalysis
- No practical attacks known against full AES
- Some theoretical attacks on reduced rounds

Attack complexity:

- Best known attacks still exponential
- Brute force remains most effective approach
- Side-channel attacks on implementations possible

Current status: AES considered secure for practical use with appropriate key sizes

Modern considerations:

- Post-quantum cryptography research
- Implementation security (timing attacks, power analysis)
- Key management and deployment practices

Summary: Block Cipher Attacks & AES

Key takeaways:

1. Key size matters - DES vulnerable due to 56-bit keys
2. Double encryption doesn’t double security - meet-in-the-middle attacks
3. S-box design is critical - poor S-boxes enable differential/linear attacks
4. Trade-offs exist between attack complexity and data requirements
5. AES represents modern design - transparent process with provable security
6. Algebraic S-box design - mathematical foundation provides strong security
7. Defense requires holistic approach - no single countermeasure sufficient

Current standard: AES provides robust security against known attacks

Questions?

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